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Activity 9: Install, Configure, and Manage Performance Monitoring tools	
1. Objectives	
Create and design a workflow that installs, configure and manage enterprise performance tools using Ansible as an Infrastructure as Code (IaC) tool.	
2. Discussion	
Performance monitoring is a type of monitoring tool that identifies current resource consumption of the workload, in this page we will discuss multiple performance monitoring tool. Prometheus Prometheus fundamentally stores all data as timeseries: streams of timestamped values belonging to the same metric and the same set of labeled dimensions. Besides stored time series, Prometheus may generate temporary derived time series as the result of queries. Source: Prometheus - Monitoring system & time series database Cacti Cacti is a complete network graphing solution designed to harness the power of RRDTool's data storage and graphing functionality. Cacti provides a fast poller, advanced graph templating, multiple data acquisition methods, and user management features out of the box. All of this is wrapped in an intuitive, easy to use interface that makes sense for LAN-sized installations up to complex networks with thousands of devices. Source: Cacti® - The Complete RRDTool-based Graphing Solution	
3. Tasks	
1. Create a playbook that installs Prometheus in both Ubuntu and CentOS. Apply the concept of creating roles. 2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.) 3. Show an output of the installed Prometheus for both Ubuntu and CentOS. 4. Make sure to create a new repository in GitHub for this activity.	

4. Output (screenshots and explanations)

install-playbook.yaml

```
HOA-9 > ! install-prometheus.yaml
1 ---
2 - name: Install Prometheus on Ubuntu
3   hosts: ubuntu_servers
4   become: true
5   gather_facts: yes
6   roles:
7     - prometheus
8   tags: ubuntu
9
10 - name: Install Prometheus on CentOS
11   hosts: centos_servers
12   become: true
13   gather_facts: yes
14   roles:
15     - prometheus
16   tags: centos
```

This Ansible playbook installs Prometheus on both Ubuntu and CentOS servers. It uses two plays, one for `ubuntu_servers` and one for `centos_servers`. The `become: true` option grants administrator privileges, and `gather_facts: yes` collects OS information. Tags (`ubuntu` and `centos`) allow for separate task execution.

main.yaml (prometheus role)

```
HOA-9 > prometheus > tasks > ! main.yaml
1 ---
2 - name: Install Prometheus on Ubuntu
3   apt:
4     name: prometheus
5     state: present
6     update_cache: yes
7   when: ansible_distribution == "Ubuntu"
8   tags: ubuntu
9
10 - name: Install EPEL repository for CentOS
11   dnf:
12     name: epel-release
13     state: present
14   when: ansible_distribution == "CentOS"
15   tags: centos
16
17 - name: Install Prometheus on CentOS
18   dnf:
19     name: prometheus
20     state: present
21   when: ansible_distribution == "CentOS"
22   tags: centos
23
24 - name: Start Prometheus service
25   service:
26     name: prometheus
27     state: started
28     enabled: yes
29   tags: prometheus
```

The Prometheus role installs and starts the monitoring tool. It first checks the `ansible_distribution` variable. On Ubuntu, it installs Prometheus using `apt`. On CentOS, it adds the EPEL repository then installs Prometheus via `dnf`. Finally, the role starts and enables the Prometheus service for automatic startup.

ubuntu_servers

```
jalene@workstation:~/CPE212_Santiago/H0A-9$ ansible-playbook install-prometheus.yaml --tags ubuntu -K
BECOME password:

PLAY [Install Prometheus on Ubuntu] *****

TASK [Gathering Facts] *****
ok: [server1]

TASK [prometheus : Install Prometheus on Ubuntu] *****
changed: [server1]

TASK [prometheus : Install EPEL repository for CentOS] *****
skipping: [server1]

TASK [prometheus : Install Prometheus on CentOS] *****
skipping: [server1]

TASK [prometheus : Start and enable Prometheus service] *****
ok: [server1]
```

```
jalene@server1: ~
jalene@server1:~$ sudo systemctl status prometheus
● prometheus.service - Monitoring system and time series database
   Loaded: loaded (/lib/systemd/system/prometheus.service; enabled; vendor pre
   Active: active (running) since Fri 2026-03-13 22:34:21 +08; 13min ago
     Docs: https://prometheus.io/docs/introduction/overview/
           man:prometheus(1)
   Main PID: 13613 (prometheus)
     Tasks: 10 (limit: 1054)
    Memory: 12.2M
       CPU: 26.136s
    CGroup: /system.slice/prometheus.service
            └─13613 /usr/bin/prometheus
```

The playbook was executed with the `ubuntu` tag, which targeted the `ubuntu_servers` group. Ansible gathered system facts and detected that the host was running Ubuntu. The Prometheus role then installed Prometheus using the `apt` package manager while skipping the CentOS-specific tasks.

The output shows that the Prometheus service is active and running, confirming that the installation was successful on the Ubuntu server.

centos_servers

```
PLAY [Install Prometheus on CentOS] *****

TASK [Gathering Facts] *****
ok: [server4]

TASK [prometheus : Install Prometheus on Ubuntu] *****
skipping: [server4]

TASK [prometheus : Install EPEL repository for CentOS] *****
ok: [server4]

TASK [prometheus : Install Prometheus on CentOS] *****
changed: [server4]

TASK [prometheus : Start Prometheus service] *****
changed: [server4]

PLAY RECAP *****
server1                : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server4                : ok=4    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

```
jalene@server4:~ — sudo systemctl status prometheus
[jalene@server4 ~]$ sudo systemctl status prometheus
● prometheus.service - Monitoring system and time series database
   Loaded: loaded (/usr/lib/systemd/system/prometheus.service; enabled; prese>
   Active: active (running) since Fri 2026-03-13 23:01:50 PST; 2min 7s ago
     Docs: https://prometheus.io/docs/introduction/overview/
           man:prometheus(1)
   Main PID: 5120 (prometheus)
      Tasks: 6 (limit: 10502)
     Memory: 47.6M (peak: 48.2M)
        CPU: 786ms
    CGroup: /system.slice/prometheus.service
            └─5120 /usr/bin/prometheus
```

The playbook was executed with the `centos` tag, targeting the `centos_servers` group. Ansible gathered system facts and identified the host as CentOS. The playbook skipped the Ubuntu installation task and proceeded to install the EPEL repository and Prometheus using the `dnf` package manager.

After the installation, the Prometheus service was started and enabled. The command `systemctl status prometheus` confirmed that the Prometheus service is active and running, indicating that the installation on the CentOS server was successful.

Reflections:

Answer the following:

1. What are the benefits of having a performance monitoring tool?

Having a performance monitoring tool helps administrators track system health and performance. It allows them to detect issues early, analyze resource usage, and ensure that servers and applications are running efficiently.

Conclusions:

This activity focuses on using Ansible to automate the installation and management of enterprise performance monitoring tools. By applying Infrastructure as Code (IaC), the workflow simplifies system deployment, ensures consistency across servers, and reduces manual configuration.